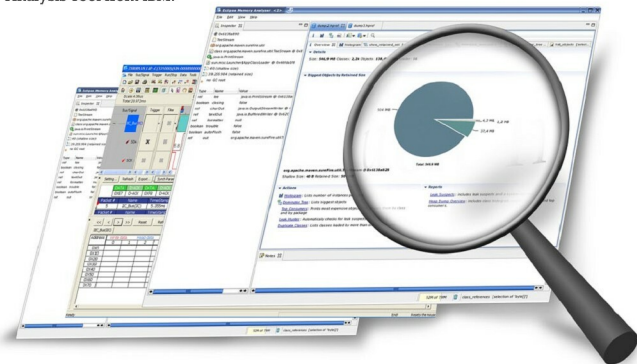


Analyse Java Memory Dump with Eclipse

This is the third and final part of the series of articles on Java heap and thread dump tools. This article covers the Eclipse Memory Analysis Tool and the Thread Dump Analysis Tool from IBM.



The default command line tools that come with Oracle JDK are useful to a limited extent but have a steep learning curve. Another problem with the default command line tools is that these are not portable across different JVMs—IBM and Oracle. In situations where the dump needs to be taken from an IBM JVM, the Oracle tools are not very helpful.

Eclipse Memory Analysis Tool

Eclipse Memory Analysis Tool (MAT) is a free tool that can be used to acquire and analyse memory dumps from both IBM and Oracle JVMs. As discussed earlier, Oracle JVM dumps are in the HPROF format and IBM dumps usually are in PHD format. The default installation of MAT can only acquire and analyse HPROF dumps. MAT can be downloaded from: www.eclipse.org/mat. Support for IBM dumps needs to be installed separately. Instructions to install the plugin to enable PHD format support are available at: <http://www.ibm.com/developerworks/java/jdk/tools/iema/>. Once this plugin is installed, MAT can be used to acquire and analyse both HPROF and PHD dumps.

Note: 1) MAT is used mainly for analysing memory dumps. Though it can open IBM's Java core file and list the threads, this information is not of much use when debugging threading issues. The heap dump files—HPROF and PHD—do not contain the threading information required to troubleshoot threading issues. 2) Both PHD and HPROF cannot be used to analyse the native memory. They are most useful in analysing objects on the heap.

Acquiring heap dump

MAT can obtain heap dumps of a locally running JVM process. To acquire a heap dump, go to *File Acquire → Heap Dump*. This option lists the various heap dump providers—Sun and IBM. It also gives the options to pass while dumping memory. Details on the options to pass can be obtained from the MAT *Help* menu. In case the dump cannot be acquired from MAT for any reason, the methods described in the first article in the series can be used to obtain the dump. The dumps obtained using command line utilities can be opened